

UK Electricity System Price Forecasting — Phase 3

Project: Day-ahead SSP forecasting · 30-min settlement period resolution | **Model:** Phase 3 Level-Shape Decomposition — June 2026 | **Horizon:** H+1 (today, 48 SPs) + H+2 (tomorrow, 48 SPs)

Problem

UK electricity System Sell Price (SSP) is highly volatile — driven by renewable intermittency, gas prices, demand seasonality, and market mechanics. Accurate day-ahead forecasts reduce balancing costs, support trading decisions, and flag high-risk periods.

Solution: Two-Stage Level-Shape Decomposition

Stage 1 — Level model (P10/P50/P90): Quantile HGBR predicts the daily average price from 85 features: price/NIV lags, day-ahead weather, wind/gas generation mix, UK CPIH inflation index, and a negative-price regime classifier.

Stage 2H+1 — Shape model (H+1): HGBR predicts how each 30-min slot deviates from the daily mean using 76 leakage-free features (fixed lags ≥ 48 SPs). At inference, BMRS WINDFOR/TSDF and Open-Meteo SP-level solar replace lag proxies with real day-ahead signals.

Stage 2H+2 — Shape model (H+2): Separate model using lag-96+ only — today's prices are not yet settled when forecasting two days ahead.

Final forecast: $\text{Level_P}[X] + \text{Shape_deviation}[\text{SP}]$ · P10/P90 bands propagate level uncertainty across all 48 SPs.

Data Sources

Source	Data	Frequency
Elexon BMRS	Settlement prices (SSP), Net Imbalance Volume	30-min, D+1 settlement
Open-Meteo	Historical weather + day-ahead forecast (3 UK sites)	30-min
Carbon Intensity	Wind %, gas % in generation mix	30-min actuals
BMRS WINDFOR+TSDF	Day-ahead wind generation & demand forecast	Per SP, D-ahead
ONS D7BT	UK CPIH index (2015=100)	Monthly

Performance

Seasonal walk-forward cross-validation — 119 days across 4 seasons (honest out-of-sample):

Season	Period	MAE	Level MAE	Shape Corr
Summer	Jul 2025	£25.4/MWh	£12.5/day	0.29
Autumn	Oct 2025	£29.6/MWh	£14.9/day	0.49
Winter	Dec 2025	£21.0/MWh	£8.4/day	0.39
Spring	Apr 2026	£33.7/MWh	£16.5/day	0.45
Aggregate	119 days	£27.4/MWh	£13.1/day	0.43

Naive lag-48 baseline: £36.3/MWh · Seasonal naive: £29.4/MWh · Phase 2 recursive: £25.4/MWh · 7-day holdout MAE £31.6 is elevated by the Jun 4 extreme renewable-oversupply event (–£70 midday prices); see reports/phase3_root_cause_analysis.md.

Key Innovations vs Phase 2

Dimension	Phase 2	Phase 3
Forecast structure	Recursive — errors compound	Non-recursive — level + shape
Horizon	H+1 only	H+1 (today) + H+2 (tomorrow)
Wind signal	None	BMRS WINDFOR (SP-level, day-ahead)
Solar signal	Daily aggregate lag	Open-Meteo SP-level day-ahead
Inflation	None	CPI-deflated training targets (ONS)

Dimension	Phase 2	Phase 3
Negative-price	None	Binary HGBR classifier → level feature
Training window	Full history	3-year rolling (reduces 2022 bias)
Evaluation	7-day holdout	4-season walk-forward CV (119 days)

Model Hyperparameters (all HGBR models)

Shared: learning_rate=0.05 · max_leaf_nodes=31 · min_samples_leaf=10 · l2_regularization=0.1 · early_stopping=True (patience=50) · max_bins=255 · loss=quantile pinball at target quantile. Parameters match Phase 2 for fair comparison; per-model tuning is a Phase 4 opportunity.

Automated Pipeline

One-click **Refresh** in the Streamlit dashboard runs: fetch prices → fetch weather → fetch generation → fetch CPI → extend dataset → rebuild features → retrain → H+1 + H+2 forecast. Dashboard panels: H+1 forecast (P10/P50/P90), H+2 forecast, verification vs actuals, price analytics, model accuracy, feature importance.

Known Limitations & Phase 4 Roadmap

- SP-level demand forecast — BMRS TSDF as a shape feature
- Specialist model for negative-price days (renewable oversupply regime)
- H+2 feature enrichment — add lag-96 NIV, weather, generation mix
- Shape model quantiles — SP-level P10/P90 calibrated intervals
- Hyperparameter tuning — separate optimisation for level vs shape models